Hyeonjun An

Ph.D. Candidate in Physics | Ulsan National Institute of Science and Technology (UNIST)
RNWK23@unist.ac.kr | hellojune726.github.io | ORCID 0009-0000-5316-6264

RESEARCH PROFILE

Experimental soft-matter physics using quantitative neutron and synchrotron X-ray imaging: polymer swelling and drying, water-vapor uptake by hygroscopic liquids, droplet freezing, and transport in porous materials.

EDUCATION

Combined M.S. and Ph.D. in Physics, UNIST Feb 2022 – Present

Expected graduation: February 2027 | Advisor: Prof. Joonwoo Jeong

B.S. in Physics and Energy Engineering, UNIST Mar 2016 – Feb 2022

Double major | Summa cum laude

Student exchange, Hong Kong University of Science and Technology (HKUST), Summer 2017.

RESEARCH EXPERIENCE

Doctoral Researcher, Soft Matter Physics Laboratory, UNIST Feb 2022 – Present

- Use H/D neutron radiography to observe hydrogel swelling and quantify solvent redistribution during PDMS swelling and drying with inverse Abel reconstruction and poromechanical modeling.
- Map water-vapor uptake and concentration gradients in hygroscopic droplets and capillaries; investigate gravity, diffusion and mixing.
- Resolve freezing fronts, polymer redistribution and brine networks using synchrotron radiography, tomography and Raman mapping.
- Measure liquid distributions and wetting fronts during imbibition of evaporative liquids in heterogeneous porous materials.

Research Placement, Pohang Accelerator Laboratory (PAL) Apr–Dec 2024

- 50 days at the PLS-II 6C BMI beamline for synchrotron X-ray imaging.

Research Internships, KRISS and UNIST 2021–2022

- Korea Research Institute of Standards and Science (KRISS), Multiscale Length Metrology Team (Jan–Feb 2022): Femtosecond laser length measurement.
- UNIST, Soft Matter Physics Laboratory (Mar–Dec 2021): Fluid dynamics in PDMS microfluidic devices.

PUBLICATIONS AND MANUSCRIPTS

1. Hyeonjun An, B. Kim, J. K. Im, S. Kim, M. W. Kim, S.-G. Kim, J.-H. Lim, K. Kim, J. Jeong. **Polymer-Regulated Freezing of Water Droplets Revealed by Synchrotron X-ray Imaging and Raman Spectroscopy.** *Journal of Colloid and Interface Science* 729, 141826 (2027; published online in 2026). DOI: 10.1016/j.jcis.2026.141826.

2. Hyeonjun An, Y. Koo, J. Cheon, P. Trtik, J. Jeong. **Condensation of the Vapor in Sessile Water Droplets and Open-Ended Water-Filled Capillaries Revealed by Neutron Radiography.** *Springer Proceedings in Physics* 348, 235–243 (2026). DOI: 10.1007/978-3-032-15003-5_27.

3. H. Yang, G. Ji, M. Choi, S. Park, Hyeonjun An, et al. **Suppressed terahertz dynamics of water confined in nanometer gaps.** *Science Advances* 10, eadm7315 (2024). DOI: 10.1126/sciadv.adm7315.

4. J. K. Im, Hyeonjun An, S.-G. Kim, J.-H. Lim, J. Jeong. **Direct X-ray visualization of bubble-mediated singularity formation in freezing droplets.** *Newton*, accepted (2026). arXiv:2511.20571.

AWARDS AND SCHOLARSHIPS

Poster Presentation Award, HANARO Symposium 2026 Nov 2026

Outstanding Poster Presentation Award Oct 2026

Korean Physical Society Fall Meeting.

First Prize for polymer swelling poster Oct 2026

10th International Topical Meeting on Neutron Radiography (ITMNR-10).

Outstanding Presentation Award in Oral Session Apr 2025; Apr 2026

Korean Physical Society Spring Meetings.

Excellence Award, DRB Space Challenge Aug 2026

Drop-tower microgravity experiment.

Misan Mountaineering Scholarship, UNIST (Fall 2026).

UNIST graduate and teaching assistant scholarships (2021–2026).

National Science and Engineering Undergraduate Scholarship, Korea Student Aid Foundation (2016–2018; 2021).

RESEARCH FUNDING

Principal Investigator, KONICOF 18 Aug 2026 – 17 Feb 2027

2026 Joint Research Program for Ph.D. Students Utilizing Overseas Research Facilities.

Quantitative neutron imaging of mass transport in soft matter with complementary X-ray imaging.

Korea Nuclear International Cooperation Foundation | International partner: Paul Scherrer Institute (PSI), Switzerland.

EXPERIMENTAL AND ANALYTICAL EXPERTISE

Neutron radiography: 70 beamtime days at ICON and NEUTRA, Paul Scherrer Institute (PSI), Switzerland, and ENF, Korea Atomic Energy Research Institute (KAERI), Korea.

X-ray imaging: 15 beamtime days at PAL PLS-II 6C BMI, 9D and 7C, plus a separate 50-day placement.

Synchrotron scattering and diffraction (SAXS/WAXD): PAL PLS-II 6D and 9A, Korea.

Analysis and modeling: H/D isotope contrast, inverse Abel reconstruction, tomography and segmentation; Flory–Rehner/Darcy and Maxwell–Stefan models.

SELECTED CONFERENCE PRESENTATIONS

International SISN Neutron Science Conference (SISNSC27) Jan 2027

Poster accepted; Florence, Italy, 18–21 January 2027.

APS DSOFT Students' Virtual Conference 10 Nov 2026

Oral presentation: "Quantifying Polymer Swelling Kinetics via Neutron Radiography."

HANARO Symposium 2026 5 Nov 2026

Polymer swelling poster (award); co-authored poster on water vapor uptake.

Korean Physical Society Fall Meeting Oct 2026

Poster: "Connectivity of Brine Networks and Tip Morphology in Freezing Saline Droplets."

10th International Topical Meeting on Neutron Radiography (ITMNR-10) Oct 2026

Two posters: "Quantifying Polymer Swelling Kinetics via Neutron Radiography" and "Direct Observation of Water Vapor Uptake in Hygroscopic Liquids by Neutron Radiography."

Korean Physical Society Spring Meetings Apr 2025; Apr 2026

Oral presentations on water vapor uptake (2025) and polymer swelling (2026).

TEACHING AND SERVICE

Teaching Assistant, UNIST (2018; 2022–2026): General Physics I and II, General Physics Laboratory I, Mathematical Physics, and Thermal and Statistical Physics.

Military service (2019–2021): Motor transport operator, 8th Fighter Wing, Republic of Korea Air Force.